



Schizophrenia

Schizophrenia is a serious psychiatric disorder that affects the way a person perceives reality, thinks, expresses emotions, and functions in daily life. It generally emerges in late adolescence or early adulthood and affects approximately one in every 345 people worldwide. The disorder can significantly affect not only a person's mental state but also their education, employment, social relationships, and family life.

The symptoms of schizophrenia are generally classified into positive, negative, and cognitive symptoms. Positive symptoms include hallucinations, such as seeing or hearing things that are not actually present, delusions, which are false beliefs that remain unchanged despite evidence to the contrary, and disorganized thoughts and behaviors. Negative symptoms include loss of interest or pleasure, reduced speech, decreased emotional expression, lack of motivation, and social withdrawal. In addition, impairments in cognitive functions such as attention, memory, problem-solving, language processing, and social judgment may occur. Cognitive impairments can cause significant difficulties in patients' ability to perform daily activities and maintain social functioning.

Schizophrenia is not thought to have a single cause. Research indicates that the interaction between genetic predisposition and environmental factors plays a role in the development of the disorder. Certain differences in brain development and abnormalities in the formation of connections between nerve cells have also been associated with the disorder. In addition, changes in neurotransmitter systems such as dopamine, glutamate, and serotonin are thought to play a role in the development of symptoms. Certain environmental factors, such as adverse conditions during pregnancy, childhood trauma, and particularly heavy cannabis use, have also been associated with an increased risk.

The primary approach to the treatment of schizophrenia involves the combined use of antipsychotic medications and psychosocial interventions. In addition, psychoeducation, family interventions, cognitive behavioral therapy, and psychosocial rehabilitation can help improve patients' daily living skills and social functioning. Clozapine is an important treatment option for treatment-resistant cases. Research on specific pharmacological treatments for cognitive symptoms is still ongoing.

Although schizophrenia can be a lifelong disorder, recovery is possible. According to the WHO, at least one-third of people with schizophrenia experience complete remission of their symptoms. Early diagnosis, regular treatment, and family and social support can increase the likelihood of an individual living an independent and meaningful life within the community. Therefore, reducing stigma and discrimination against people with schizophrenia is as important as treatment itself.

References:

- 1) World Health Organization. (2025, October 6). Schizophrenia. <https://www.who.int/news-room/fact-sheets/detail/schizophrenia>
- 2) Javitt DC. Cognitive Impairment Associated with Schizophrenia: From Pathophysiology to Treatment. *Annu Rev Pharmacol Toxicol*. 2023 Jan 20;63:119-141. doi: 10.1146/annurev-pharmtox-051921-093250. Epub 2022 Sep 23. PMID: 36151052.
- 3) Cummings MA, Arias AW, Stahl SM. What is the neurobiology of schizophrenia? *CNS Spectr*. 2024 Oct 30;30(1):e13. doi: 10.1017/S1092852924000518. PMID: 39473188; PMCID: PMC13064734.

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